Alan Tseng

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SUMMARY

Firmware engineer with experience on both the managed-silicon side (Android SoC/AP platform integration) and the management-controller side (OpenBMC device enablement and firmware debugging on ASPEED BMCs). Comfortable at the register and bit-field level – device enablement and firmware crash debugging with gdbserver, core dump analysis, perf, and uftrace. M.Sc. from the Graduate Institute of Networking and Multimedia, National Taiwan University. COSCUP speaker on OpenBMC debugging (2025) and agentic AI for firmware (2026); builds AI agents to automate platform engineering work.

SKILLS

Languages	C, C++, Python, Go, Shell, JavaScript / TypeScript
Systems & Kernel	firmware / core dump debugging, Android AOSP, device tree, drivers (I2C, SPI, GPIO, ADC, UART), systemd, D-Bus, performance analysis (perf, ftrace, uftrace), GDB / gdbserver, core dump analysis
Firmware & Embedded	OpenBMC, Yocto, Buildroot, IPMI, Redfish, BMC firmware development, sensor enablement
AI & Automation	LLM agent frameworks, RAG (embedding + vector store retrieval), tool-use / function calling, agent harness design
Infrastructure & Cloud	Kubernetes (kubeadm, Helm), Docker, PXE / DHCP / TFTP provisioning, CI/CD pipelines, container image optimization
Web & Tooling	WebGPU, WebAssembly (WASM), Web APIs, browser-based benchmarking, REST API design, pytest

EXPERIENCE

Project System Integration Engineer – Android Platform Sep. 2025 – Present
MediaTek, Taipei, Taiwan

- Maintain branch, repo, CI, and image pipelines and own a driver module for the Amazon tablet line; trace AOSP build failures, runtime logs, and benchmark regressions to root cause and drive fixes across software, firmware, and QA teams.
- Built a browser-based benchmarking suite (WebGPU, WebAssembly, Web APIs) to estimate SoC and GPU performance on locked commercial devices with no root access, which made cross-vendor comparison possible on production tablet SKUs.
- Ran thermal and low-power experiments on the tablets and correlated benchmark scores with throttling behavior and power-rail telemetry. The resulting reports fed into product and platform decisions.
- Wrote a set of internal AI agent skills that automate repetitive Android platform work. The system-upgrade skill pulls target changes from Gerrit and runs local verification. A separate build-and-boot skill fetches the code, builds the image, flashes the device, and confirms it boots all the way to the home screen.
- Built a skill that integrates with internal Teams and Outlook to notify the relevant code owners automatically, removing a manual coordination step.
- As driver module owner, systematized the new-chip porting workflow (config, device tree nodes, register base updates, build wiring) into an internal AI agent skill, making repeat ports reproducible.
- These skills earned an internal vAward. The design kept security first, so engineering teams could adopt AI agents without exposing internal source or credentials.

Embedded Systems Engineer – BMC Firmware Dec. 2024 – Sep. 2025
Wiwynn, New Taipei City, Taiwan

- Developed and debugged OpenBMC firmware on Yocto and Buildroot hyperscale server platforms, working on sensors, services, and management interfaces.
- Developed device-specific handling in the PMBus driver path – I2C/SMBus register access and bit-field decoding per vendor datasheet – to expose correct hwmon/sysfs sensor values for Amazon platform requirements.
- Ported and validated device tree, I2C, GPIO, ADC, systemd, and D-Bus interfaces, and validated IPMI and Redfish management endpoints against datacenter customer specs.
- Built a cross-debugging workflow with gdbserver, core dump analysis, and live-attach to triage BMC firmware crashes and service failures on production servers. Presented it as a public talk at COSCUP 2025.
- Wrote Python and pytest validation tools that cut manual sanity-check time and caught regressions before customer hand-off.

Server Software Engineer – Provisioning Automation Sep. 2024 – Dec. 2024
Pegatron, Taipei City, Taiwan

- Designed and built Kubernetes-based PXE provisioning for rack-scale ARM and x86 server validation, automating Windows and Linux OS deployment with kubeadm, Helm, Docker, DHCP, and TFTP.
- Wired low-level networking (ARP, DHCP, TFTP) into the pipeline and added stage-by-stage logging so boot failures could be diagnosed without a physical console.
- Cut container image and repository size from 663 MB to 2.4 MB (about 99%) by restructuring base images, dropping redundant artifacts, and rewriting Dockerfiles as multi-stage builds.

Software Engineer Intern Jul. 2022 – Nov. 2022
LINE Corporation, Taipei City, Taiwan

- Cut backend API development time by about 20% by introducing Swagger-driven code generation and standardizing API specs across the B2B platform team.

EDUCATION

M.Sc., Graduate Institute of Networking and Multimedia, College of Electrical Engineering & Computer Science, National Taiwan University Aug. 2021 – Feb. 2024 | GPA: 4.12/4.3
B.S., Computer and Communication Engineering, Ming Chuan University Sep. 2017 – Jun. 2021 | GPA: 3.99/4.0

SELECTED PROJECTS

- **CPU-Side Low-Bit LLM Inference Analysis (T-MAC / BitNet)** – Profiled inference performance using `perf` and `uftrace`; analyzed IPC, cache and TLB misses, page faults, and function-level execution behavior to identify optimization targets.
- **Linux Kernel Internals & Open-Source Contributions** – Contributed course materials, including eBPF use cases in cloud-native environments, profiling, and memory management; contributed to *Operating System in 1000 Lines*. wiki
- **Adaptive Driving Beam System** – Designed a nighttime vehicle-detection and adaptive-beam system using CycleGAN, YOLO, statistical analysis, and SVM; published in CVGIP 2020.
- **Technical Writing** – Maintain a TIL and blog covering embedded systems, Linux debugging, AI-assisted development, and platform engineering workflows. TIL | Blog

TALKS, HONORS AND LEADERSHIP

- **Talk** – *Agentic AI for Firmware Development*, COSCUP 2026. How LLM agents fit into firmware and embedded work: understanding code, assisting debugging, and using tools safely inside engineering pipelines.
- **Talk** – *Debugging OpenBMC Firmware*, COSCUP 2025. Cross-debugging on production BMC platforms with gdbserver, core dump analysis, and live-attach.
- **Honors** – Research Excellence Award (NTU); Intel Andy Grove Scholarship; College Student Research Scholarship from Taiwan's Ministry of Science and Technology.
- **Leadership** – Lead, Google Developer Student Club; Leader, ACM Competitive Programming Club; volunteer staff at SITCON and COSCUP community events.